



The paradoxical relationship between students' non-cognitive factors and mathematics & science achievement using TIMSS 2015 dataset

Minge Chen^{*}, Dirk Hastedt

International Association for the Evaluation of Educational Achievement (IEA), the Netherlands

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ABSTRACT

This study aims to fill in the gap of the accumulating literature on the paradoxical relationship between students' non-cognitive variables and their academic achievement. Previous studies either focus on a single item or didn't check the measurement invariance of the non-cognitive variables. This study first checked the measurement invariance of two important non-cognitive variables including students' confidence and interest using TIMSS 2015 data and found that both scales are metric invariant and thus the comparison of the correlation coefficients across different countries are possible. Correlational analysis indicates that the paradox exists at country level for all countries and at class level only for a few countries.

1. Introduction

Both students' academic self-concept (ASC) and their interest in school subjects are non-cognitive factors that function as a driving force for students' school engagement and their academic performance (Köller, Baumert, & Schnabel, 2001; Pintrich & Schunk, 1996; Viljaranta, Tolvanen, Aunola & Nurmi, 2014).

1.1. Academic self-concept (ASC)

Students' ASC refers to students' perceptions and attitudes toward their capacities and performance (Areepattamannil & Freeman, 2008; Lent, Brown, & Gore, 1997). It is an essential component of studies on school psychology as it impacts students in a variety of aspects including their motivation, school engagement and achievement, course/major selection in schools and even their future career path and accomplishments (Marsh, Abduljabbar, Parker, Morin, Abdelfattah & Nagengast, 2014). The association between students' ASC and their academic performance has been studied by many scholars (Areepattamannil & Freeman, 2008; Cokley, 2007; Marsh, 1990; Marsh, Trautwein, Lüdtke, Köller & Baumert, 2005; Valentine, DuBois, & Cooper, 2004; Viljaranta et al., 2014), and it has been found that students with higher ASC tend to perform better in school (Cokley & Patel, 2007; Marsh, 1990; Marsh et al., 2005; Pintrich & Schunk, 1996). In addition, researchers have identified a reciprocal relationship between students' ASC and their academic performance. Specifically, students' prior achievement in

school can positively affect their ASC, subsequently a positive ASC is associated with better academic performance even after controlling for previous performance (Marsh et al., 2014). Therefore, students' ASC can both be a cause and effect of their academic achievement (Eccles et al., 1983; Marsh et al., 2005; Viljaranta et al., 2014).

Students' ASC is not only shaped by their previous performance but also by the achievement of their counterparts within the same class or school (Marsh et al., 2014). Adopting different frame of reference can lead to disparate ASC for the equally able students measured by objective standards (e.g., standardized tests). For example, students enrolled in selective schools or gifted education programs tend to underestimate their true academic capacity if they compare themselves with their classmates who are mostly exceptional students; however, if the same students were placed in less selective or even low-performing environment, their self-concept would be more positive. This contextual effect on students' ASC is often referred to as "Big-Fish-Little-Pond-Effect" (BFLPE) (Marsh & Parker, 1984; Marsh et al., 2014).

1.2. Academic interest

Students' general academic interest is regarded as a salient intrinsic motivational determinant for success in school (Köller et al., 2001). Normally, people's motivation to exercise something can be either intrinsic or extrinsic. Intrinsic motivation derives from the subject or the activity itself (Eccles et al., 1998; Gottfried et al., 2001). Specifically in school, it is the pleasure or enjoyment of the learning process that

^{*} Corresponding author.

E-mail address: chenmi613@outlook.com (M. Chen).

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enables the students to pursue satisfactory performance on a certain subject. Students with strong intrinsic motivation toward schooling tend to exhibit the following characteristics including initiative orientation, curiosity, persistence, and confidence in facing challenging tasks (Gottfried et al., 2001). Accumulated research evidence have indicated that students' academic interest is positively associated with their performance (Köller et al., 2001; Aunola, Leskinen, & Nurmi, 2006; Schiefele, 1992) and similar to ASC, this association is also reciprocal (Aunola et al., 2006; Viljaranta et al., 2014).

1.3. Paradoxical relationship in the field of international large-scale assessments

The paradoxical relationship between non-cognitive factors, especially ASC and students' academic performance using International Large-Scale Assessments (ILSAs) has been of interest of educational researchers for nearly two decades (Liou, 2014; Marsh et al., 2014; Marsh, Abduljabbar, Parker, Morin, Abdelfattah & Nagengast, 2015; Shen & Pedulla, 2000; Shen & Talavera, 2003; Shen & Tam, 2008; Seaton, Marsh, & Craven, 2009; Wang, 2015; Wang & Bergin, 2017; Wang, 2020). Specifically, the paradoxical relationship refers to a phenomenon that at the individual level, there is a positive association while at a higher aggregated or ecological level, the association turns to be negative.

The most frequently used ILSAs data in such paradox studies are either from Trends in International Mathematics and Science Study (TIMSS) or from the Organization for Economic Co-operation and Development Program for International Student Assessment (PISA).

1.4. Studies using TIMSS data

Shen and his colleagues' research is among the earliest paradox studies using ILSAs datasets. Shen & Pedulla (2000) analyzed TIMSS 1995 data and found out that at individual level, there is a positive association between students' mathematics and science achievement and their self-perception in the two subjects (measured by two items: students' self-perceived competence in the subjects and students' perceived easiness of the subjects); however, at country level, the association turns to be negative. And this paradox exists for all age cohorts (3rd, 4th, 7th, and 8th graders). In Shen's following studies (Shen, 2002; Shen & Talavera, 2003; Shen & Tam, 2008), he and his colleagues continued to analyze TIMSS 1999 and TIMSS 2003 data and claimed consistent findings for 8th graders for both mathematics and science. Shen and Talavera (2003) also adopted the school as the unit of analysis to check the student-school level paradox and identified a mixed picture: the paradox is not present for how much students like the subjects and how well they think they are doing with the subjects but quite pronounced for students' perceived easiness of the two subjects. They proposed that the rigor of the curriculum leads to such paradox. Specifically, less rigorous curriculum and loose scoring standards lead students to falsely believe that their academic capacity is high or the subject matter is easier than it actually is (Shen & Pedulla, 2000; Shen, 2002; Shen & Talavera, 2003; Shen & Tam, 2008).

The series of Shen's studies progressed along the time by including more educational systems, analyzing more self-perception items, controlling covariates (Shen, 2002; Shen & Talavera, 2003; Shen & Tam, 2008), and taking the complex survey design into consideration (Shen & Talavera, 2003; Shen & Tam, 2008). It is noted that these studies use single items rather than multi-item scales to measure student self-concept. This is probably due to the fact that the self-perception scale is only available till 2011 for TIMSS. It is a long debated topic as to whether multi-item measure outperforms single item measure (Bergkvist & Rossiter, 2007; Diamantopoulos, Sarstedt, Fuchs, Wilczynski & Kaiser, 2012; Sarstedt & Wilczynski, 2009). Generally, using single item to measure a construct is regarded as skeptical or even a grave error due to its lack of reliability and validity (Sarstedt &

Wilczynski, 2009). Another limitation of Shen's studies is that the majority of them focused on eighth graders (Shen, 2002; Shen & Talavera, 2003; Shen & Tam, 2008).

Later, Marsh and his colleagues used TIMSS 2007 data to test the cross-cultural generalizability of Big Fish Little Pond Effect (BFLPE) and tried to earn its place as a pan-human theory (Marsh et al., 2014; Marsh et al., 2015). BFLPE is a theory proposed by Marsh (1987). Basically, BFLPE suggests that students tend to use their peers' ability as a frame of reference to form their own academic self-concept and thus students with the same capacities tend to underestimate themselves when attending schools where students' average performance is high while overestimate themselves when attending schools where students' average performance is low. Compared with Shen's studies, Marsh' studies are methodologically more advanced. They adopted Doubly Latent Models to check the BFLPE across countries. Such method is a combination of Structural Equation Modeling (SEM) and Multilevel Modeling (MLM) to account for measurement and sampling error (Marsh et al., 2009). Both studies conducted measurement invariance of the self-concept construct and their analyses supported that the self-concept construct reached metric invariant, indicating that the association between the latent variable and the observed items are the same across groups. However, the number of groups involved in the two studies are small (two countries and 26 countries, respectively). It is noted that Marsh et al. (2015) study also include primary school students (fourth graders) and is the first study to use class as the unit of analysis. Wang (2015) is another study to analyze BFLPE using TIMSS 2007 data. It also adopted Doubly Latent Model and the measurement invariance of the self-concept construct across 49 countries was tested at both within level and between level.

Only two studies have been found using TIMSS 2011 data to explore BFLPE. One is Liou (2014), which only focused on Taiwan while another one is Wang and Bergin (2017), which examined BFLPE for eighth grade mathematics across 59 countries and regions. Wang and Bergin (2017) suggested that BFLPE existed in 58 out of 59 countries. The measurement invariance of students' math self-concept was tested, however, only configural measurement invariance has been reached. Therefore, the association between students' mathematics achievement and their mathematics self-concept could only be analyzed in each country and both the magnitude and the direction of the association could not be compared across countries. Besides, compared with Marsh's studies, Wang used 3 items ("I usually do well in mathematics"; "mathematics is not one of my strengths"; and "I learn things quickly in mathematics") to measure students' mathematics self-concept while Marsh used 4 items, the additional item is "mathematics is more difficult for me than for many of my classmates". Thus, there is inconsistency on how to measure the self-concept construct among the paradox studies.

The latest paradox study using the most updated TIMSS data (TIMSS 2015) is Wang (2020). This study is an extension of Wang and Bergin (2017) and include also fourth graders.

1.5. Studies using PISA data

Paradox studies have also been conducted using Program for International Student Assessment (PISA) data. Marsh and Hau (2003) used PISA 2000 data to analyze BFLPE across 26 industrialized western countries and obtained consistent findings with TIMSS data that the school average achievement (including mathematics, science, and reading) is negatively associated with students' self-concept. Later studies using PISA 2003 data (Seaton, Marsh, & Craven, 2009; Seaton et al., 2010) increased the sample size of the educational systems to include both developed and developing countries. The findings of these studies support the universality of BFLPE. However, two limitations exist in those studies. First, the invariance level of the confidence scale was not checked beforehand; second, the cultural generalizability of BFLPE is based on school-aggregated results since the sampling unit in PISA is the whole school; which can cause concern because BFLPE is

more accurate if the frame of reference is more local (e.g., average performance of classmates vs. average performance of students in the same school). (Marsh et al., 2015; Zell & Alickie, 2009).

After examining the body of literature on the paradox under the context of ILSAs, we can suggest that the paradoxical association between students' self-concept and their academic performance is common at different levels (class, school, and country levels) across different cultural groups. However, there is still room for further studies specifically in the following areas:

1.6. The construct of the self-concept should be consistently measured across studies

In Shen's studies, self-concept is measured by single items; in Marsh's studies, self-concept is measured by four items (see Table 2); and in Wang's studies, self-concept is measured by three items (see Table 2). If the self-concept is not measured by the same items at the first place, we should be very careful to compare the findings from these studies.

1.7. The construct of the self-concept should be invariant across different groups in each study

Measurement non-invariance is listed as one of the major pitfalls in cross cultural studies (Rottig, 2009). If the construct in one cultural group differs from that of another cultural group, the analysis results based on these non-invariant constructs could be "at best ambiguous and at worst erroneous" (Steenkamp & Baumgartner, 1998, p. 78). Specifically for the paradox studies, if the construct is different conceptually (e.g., the construct refers to different underlying factor or is composed of different items), there is no common ground to discuss any observed differences of the construct; if the construct is the same conceptually but different numerically (e.g., the association between the items and the construct is different across groups), statistical analysis could still be done but only within each group and cannot be compared with each other. For paradox studies, two kinds of measurement invariance of the self-concept scale should be taken into consideration. One is multilevel measurement invariance, which indicates that the self-concept scale should be measured the same both at student level and ecological levels (including class level, school level, and country level); another type of measurement invariance is that at each level, the scale should be invariant across different groups (e.g., countries, regions, or educational systems). In the existing literature, the issue of measurement invariance of the self-concept scales has just been raised up and taken care of in

Table 1
Measured Items for "Students Like Learning Mathematics and Science" for Fourth/Eighth Graders.

How much do you agree with these statements about learning mathematics?	
ASBM01A/BSBM17A	I enjoy learning mathematics
ASBM01B/BSBM17B	I wish I did not have to study mathematics
ASBM01C/BSBM17C	Mathematics is boring
ASBM01D/BSBM17D	I learn many interesting things in mathematics
ASBM01E/BSBM17E	I like mathematics
ASBM01F/BSBM17F	I like any school work that involves numbers
ASBM01G/BSBM17G	I like to solve mathematics problems
ASBM01H/BSBM17H	I look forward to mathematics lessons/class
ASBM01I/BSBM17I	Mathematics is one of my favorite subjects
How much do you agree with these statements about learning science?	
ASBS04A/BSBS21A	I enjoy learning science
ASBS04B/BSBS21B	I wish I did not have to study science
ASBS04C/BSBS21C	Science is boring
ASBS04D/BSBS21D	I learn many interesting things in science
ASBS04E/BSBS21E	I like science
ASBS04F/BSBS21F	I look forward to learning science in school
ASBS04G/BSBS21G	Science teaches me how things in the world work
ASBS04H/BSBS21H	I like to do science experiments
ASBS04I/BSBS21I	Science is one of my favorite subjects

recent years. This is probably due to the fact that there is no self-concept scale in TIMSS or PISA dataset until recently. Before that, single items were used as self-concept measure. Marsh and his colleagues' BFLPE studies (Marsh et al., 2014, 2015) are pioneering studies which have checked the invariance of the self-concept scale before further examining the paradoxical association between this scale and students' academic achievement. However, these studies didn't check multilevel measurement invariance of self-concept scale and the countries/regions involved in the invariance test is limited. Marsh et al. (2014) has only two countries (United States and Saudi Arabia) involved and Marsh et al. (2015) has 13 countries involved. Wang & Bergin (2017) and Wang (2020) checked both the multilevel measurement invariance of self-concept within each country/region and the invariance of self-concept across different countries/regions, and the countries/regions involved is Wang's studies are much more than Marsh's study. However, it is noted that in Wang's studies, the self-concept scale is only composed of three items. More details on different invariance levels and the appropriate analysis associated with each level is discussed in the methodology section.

1.8. Purpose of the study

Taken all these into consideration, this study aims to first check the measurement invariance of the non-cognitive variables including students' confidence in mathematics and science, and students' interest in mathematics and science across different countries/regions before proceeding with further analysis. It is noted that students' interest in mathematics and science has never been investigated for its invariance in the paradox studies. Next, the study conducts correlational analysis to explore whether the paradox exists using TIMSS 2015 data. If such paradox exists, is it because of East Asian countries? Previous international assessments have identified two groups of high achieving countries/regions: East Asian Countries and European Countries (Lee, 2009; Stankov, 2010; Wilkins, 2004). However, compared with their European counterparts, who are also high achievers, East Asian students normally scored low in self-concept. This phenomenon has become known as the *East Asian Learner Paradox* (Stankov, 2010). Some researchers are suspicious of the actual existence of the *East Asian Learner Paradox*. Some argued that the discrepancy of students' self-evaluation of these non-cognitive variables is the result of measurement issue that the non-cognitive indicators are only measured by very few items or even a single item (Stankov, 2010). Some other researchers proposed that the achievement gap observed between East Asian students and European students are exaggerated and thus does not represent the real differences of school performance between the two cultural groups (Hogan, 2008). Their reasoning is that East Asian pedagogy emphasizing memorization and repetition helps to produce high performers for TIMSS, but that is not the case for PISA since PISA requires more problem-solving skills for real life (Hogan, 2008). Despite the challenges of the existence of the *East Asian Learner Paradox*, Stankov (2010) suggests that the Confucius culture shared among the East Asian countries could be the driving force behind the *East Asian Learner Paradox*. The Confucius culture emphasizes academic achievement, competition, and collectivism and thus students tend to be more anxious and self-doubting (Wilkins, 2004; Lee, 2009; Stankov, 2010).

Therefore, we would like to explore whether the paradox is unique to East Asian countries or universal? However, it is noted that, this study on the paradox is only based on bivariate correlational analysis and thus exploratory. Hopefully, the results provided in this study could serve as a starting point to further explore the paradoxical association between non-cognitive factors and students' academic achievement using more advanced methodology like the doubly latent models.

Table 2

Measured Items for “Students Confidence in Mathematics and Science” for Fourth/Eighth Graders.

How much do you agree with these statements about learning mathematics?	
ASBM03A/ BSBM19A	I usually do well in mathematics ^{*,†}
ASBM03B/ BSBM19B	Mathematics is harder/more difficult for me than for many of my classmates*
ASBM03C/ BSBM19C	I am just not good at mathematics/Mathematics is not one of my strengths ^{*,†}
ASBM03D/ BSBM19D	I learn things quickly in mathematics ^{*,†}
ASBM03E/ BSBM19E	Mathematics makes me nervous
ASBM03F/ BSBM19F	I am good at working out difficult mathematics problems
ASBM03G/ BSBM19G	My teacher tells me I am good at mathematics
ASBM03H/ BSBM19H	Mathematics is harder for me than any other subject
ASBM03I/ BSBM19I	Mathematics makes me confused
How much do you agree with these statements about learning science?	
ASBS06A/ BSBS23A	I usually do well in science
ASBS06B/ BSBS23B	Science is harder/more difficult for me than for many of my classmates
ASBS06C/ BSBS23C	I am just not good at science/Science is not one of my strengths
ASBS06D/ BSBS23D	I learn things quickly in science
ASBS06E/ BSBS23E	My teacher tells me I am good at science
ASBS06F/ BSBS23F	Science is harder for me than any other subject
ASBS06G/ BSBS23G	Science makes me confused
ASBS06H/ BSBS23H	NA/I am good at working out difficult science problems

Note.

*indicates the items used by Marsh to form self-concept construct (Marsh et al., 2014, 2015).

†indicates the items used by Wang to form self-concept construct (Wang, 2015, 2020; Wang & Bergin, 2017).

2. Data and methodology

2.1. Data

The data used in this study comes from the TIMSS 2015 dataset. TIMSS is an ILSA of mathematics and science for fourth and eighth graders which has been conducted every four years since 1995 (Mullis, Martin, Foy & Hooper, 2016). TIMSS 2015 includes data collected from 57 countries and 7 benchmarking entities, which are additional educational systems within a country. The sample size of the study is above 580,000 students (Mullis et al., 2016).

2.2. Variables

Non-cognitive scales included in this paper are: *Students Like Learning Mathematics (fourth and eighth graders)*, *Students Like Learning Science (fourth and eighth graders)*, *Students Confidence in Mathematics (fourth and eighth graders)*, and *Students Confidence in Science (fourth and eighth graders)*. The items measured by these scales appear below in Table 1 and Table 2. As shown in Table 2, more items are included into the self-concept scale in this study than in previous studies (Marsh et al., 2014, 2015; Wang, 2015, 2020; Wang & Bergin, 2017).

2.3. Measurement invariance analysis

Measurement invariance describes the comparability of a scale across different groups. In the CFA framework, a hierarchy of

measurement invariance is defined as configural invariance to metric invariance/weak factorial invariance (Meredith, 1993) to scalar invariance/strong factorial invariance (Meredith, 1993), where each type of invariance allows for greater comparability of the scale among groups (Milfont & Fischer, 2010). In the configural model, we allow both the factor loadings and intercepts to be freely estimated across countries. If the scale reaches configural invariance, we can conclude that the configuration of the scale is the same for each group and thus the scale can, at least, be compared conceptually across groups. In the metric model, we constrained the factor loadings to be the same across groups, while allowing intercepts to be freely estimated. When a scale reaches metric invariance, we can assume that not only the scale is composed of the same items, but also the association between the scale and items are the same across countries. To put it another way, one unit of change in the items will result in the same units of change in the scale for all groups. When metric invariance is reached, correlation and regression analysis are possible for such scales. In the scalar model, we constrain both the factor loadings and the intercepts to be the same across countries. If a scale exhibits scalar invariance, not only are associations between the scale and items the same for groups, but also the intercepts. Therefore, the mean values of such scales are comparable across groups. Therefore, a scale has to be at least metric invariant in order to compare the correlation or regression coefficients (both the magnitude and the direction) across different groups (either across different education systems or across different levels). Specifically for paradox studies, metric invariance is a prerequisite since we need to compare the direction of the association at different levels.

However, the hierarchy of measurement invariance discussed above is developed based on continuous CFA modeling wherein the items to measure the unobserved factors are continuous variables. In our study, the items of all the eight non-cognitive scales are four-point likert scale (“Agree a lot”, “Agree a little”, “Disagree a little”, and “Disagree a lot”), therefore, categorical CFA seems to be a more appropriate modeling approach. The measurement invariance of categorical CFA is a little bit different from that of continuous CFA and the literature on the extension of categorical CFA to multiple population is still limited and relatively new (Millsap & Yun-Tein, 2004; Muthén & Asparouhov, 2002; Rutkowski, Svetina, & Liaw, 2019; Svetina, Rutkowski, & Rutkowski, 2020; Wu & Estabrook, 2016). Similar to continuous CFA modeling, configural model is first tested, followed by equal slopes model, and then followed by more restrictive model: equal slopes and equal thresholds. The three models are analogous to configural, metric, and scalar models within continuous CFA modeling (Rutkowski & Svetina, 2017). Multiple Group Confirmatory Factor Analysis (MG-CFA) was conducted using Mplus 8 to check the measurement invariance of the eight non-cognitive scales in the study.

We compared the three models through both the chi-square test and through comparing model fit indices (e.g., CFI, TLI, RMSEA, and WRMR). The chi-square test, in our case, is not very informative since it is very sensitive to sample size while ILSA studies usually have big sample size (Bagozzi, 1977; Bentler & Bonett, 1980; Rutkowski & Svetina, 2014); therefore, we only referred to the absolute change of the model fits to decide which measurement invariance level the scale has achieved. To speak in another way, if the model does not deteriorate significantly after being imposed with more constraints, we would claim that the scale reaches a higher level of measurement invariance.

2.4. Correlational analysis

For scales that met at least metric invariance, correlational analyses were further conducted at individual level and ecological level (country & class). Due to the complex survey design of TIMSS dataset, we used IDB analyzer to conduct such analyses. The IDB Analyzer offers the advantages of accounting for specifics of the TIMSS study, that are weights, replication, and plausible values. In the ecological level analysis, the country mean/class mean is the unit of analysis. Then we could

Table 3

Measurement Invariance for Students Like Learning Mathematics in TIMSS 2015.

	Grade 4				Grade 8			
	CFI	TLI	RMSEA	WRMR	CFI	TLI	RMSEA	WRMR
Configural	0.979	0.972	0.095	24.517	0.980	0.974	0.110	25.596
Metric	0.979	0.972	0.095	24.518	0.980	0.974	0.110	25.597
Scalar	0.978	0.985	0.070	56.889	0.980	0.987	0.078	57.791
	ΔCFI	ΔTLI	$\Delta RMSEA$	$\Delta WRMR$	ΔCFI	ΔTLI	$\Delta RMSEA$	$\Delta WRMR$
Metric VS Configural	0	0	0	0.001	0	0	0	0.001
Scalar VS Metric	-0.001	0.013	-0.025	32.371	0	0.013	-0.032	32.194

Table 4

Measurement Invariance for Students Like Learning Science in TIMSS 2015.

	Grade 4				Grade 8			
	CFI	TLI	RMSEA	WRMR	CFI	TLI	RMSEA	WRMR
Configural	0.971	0.961	0.107	30.391	0.965	0.953	0.137	32.669
Metric	0.971	0.961	0.107	30.393	0.964	0.953	0.137	32.67
Scalar	0.978	0.985	0.066	51.748	0.979	0.986	0.076	51.312
	ΔCFI	ΔTLI	$\Delta RMSEA$	$\Delta WRMR$	ΔCFI	ΔTLI	$\Delta RMSEA$	$\Delta WRMR$
Metric VS Configural	0	0	0	0.002	-0.001	0	0	0.001
Scalar VS Metric	0.007	0.024	-0.041	21.355	0.015	0.033	-0.061	18.642

Table 5

Measurement Invariance for Students Confidence in Math in TIMSS 2015.

	Grade 4				Grade 8			
	CFI	TLI	RMSEA	WRMR	CFI	TLI	RMSEA	WRMR
Configural	0.987	0.977	0.053	13.454	0.985	0.974	0.073	16.067
Metric	0.986	0.976	0.055	13.834	0.985	0.973	0.074	16.222
Scalar	0.944	0.957	0.073	50.494	0.959	0.968	0.081	53.052
	ΔCFI	ΔTLI	$\Delta RMSEA$	$\Delta WRMR$	ΔCFI	ΔTLI	$\Delta RMSEA$	$\Delta WRMR$
Metric VS Configural	-0.001	-0.001	0.002	0.380	0	-0.001	0.001	0.155
Scalar VS Metric	-0.042	-0.019	0.018	36.66	-0.026	-0.005	0.007	36.83

Table 6

Measurement Invariance for Students Confidence in Science in TIMSS 2015.

	Grade 4				Grade 8			
	CFI	TLI	RMSEA	WRMR	CFI	TLI	RMSEA	WRMR
Configural	0.993	0.987	0.046	9.263	0.991	0.983	0.062	10.384
Metric	0.993	0.986	0.048	9.548	0.989	0.978	0.07	11.652
Scalar	0.952	0.968	0.073	49.834	0.949	0.962	0.093	55.713
	ΔCFI	ΔTLI	$\Delta RMSEA$	$\Delta WRMR$	ΔCFI	ΔTLI	$\Delta RMSEA$	$\Delta WRMR$
Metric VS Configural	0	-0.001	0.002	0.285	-0.002	-0.005	0.008	1.268
Scalar VS Metric	-0.041	-0.018	0.025	40.286	-0.040	-0.016	0.023	44.061

compare the individual level correlation coefficient with the aggregated correlation to examine whether educational paradox exists.

For most of the countries participating in TIMSS 2015, only few classes were sampled to represent a school (Marsh et al., 2015), we assume that the school level and class level analyses would be very similar. Therefore, we only did class level correlation analysis.

3. Results

3.1. Measurement invariance on ASC scale

Table 3 to Table 6 list the model fits and the change of the fits between less restrictive and more restrictive models for measurement invariance test for categorical CFA modeling. It is noted that for the four self-concept scales, the configural model didn't perform well, therefore, some residual correlations were added to improve the model. For *Students Confidence in Mathematics (Fourth and Eighth Graders)*, "I usually do well in mathematics", "I learn things quickly in mathematics", "I am good at working out difficult mathematics problems", and "My teacher tells me I am good at mathematics" are correlated with each other. For

Students Confidence in Science (Fourth Graders), "I usually do well in science", "I learn things quickly in science", and "My teacher tells me I am good at science" are correlated. For *Students Confidence in Science (Eighth Graders)*, "I usually do well in science", "I learn things quickly in science", "My teacher tells me I am good at science", and "I am good at working out difficult science problems" are correlated.

According to Rutkowski and Svetina (2014), for ILSAs with large number of groups, a less strict criterion for measurement invariance should be adopted. Therefore, we decided to use the following criteria to determine the invariance level of the scales: $\Delta CFI \geq -0.2$ for metric invariance and $\Delta CFI \geq -0.1$ for scalar invariance; $\Delta TLI \geq -0.15$ for both metric and scalar invariance; $\Delta RMSEA \leq 0.03$ for metric invariance and $\Delta RMSEA \leq 0.01$ for scalar invariance; $\Delta SRMR \leq 0.03$ for metric invariance and $\Delta SRMR \leq 0.01$ for scalar invariance. For configural models, CFI and TLI should exceed .9, RMSEA should be less than .08 and WRMR should be less than .09. It is suggested that WRMR should not be used in isolation but rather be used together with CFI and TLI. If there is contradiction between model fit indices, CFI and TLI should be preferred (Rutkowski and Svetina, 2014).

The results indicate that all the scales meet at least metric invariance

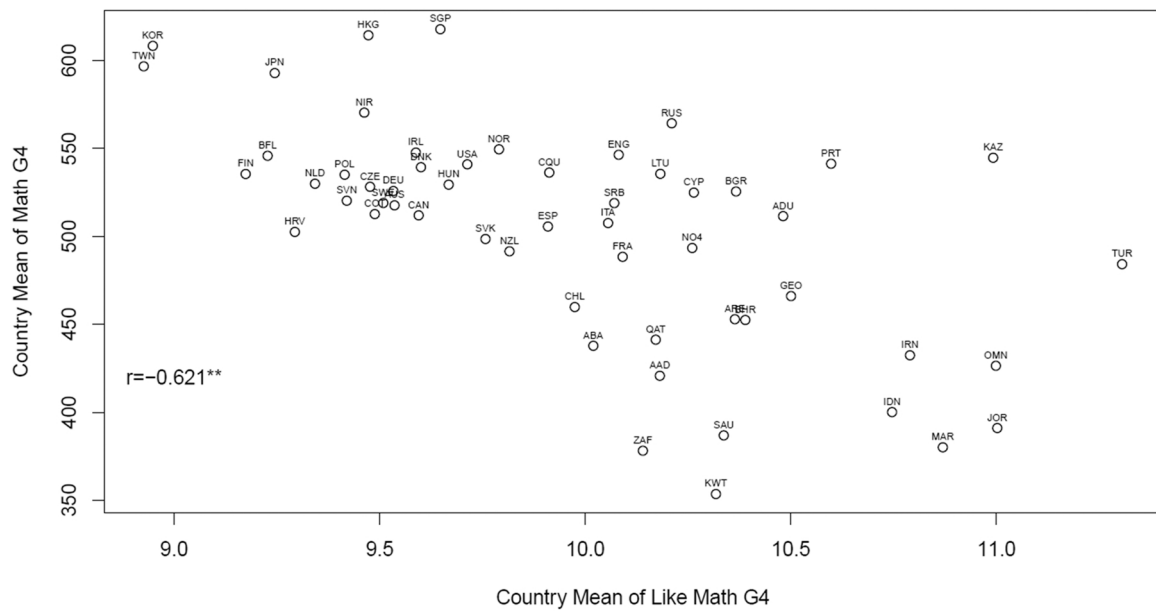


Fig. 1. Country Aggregated Correlation between Students Like Learning Mathematics and Mathematics Performance for Fourth Graders.

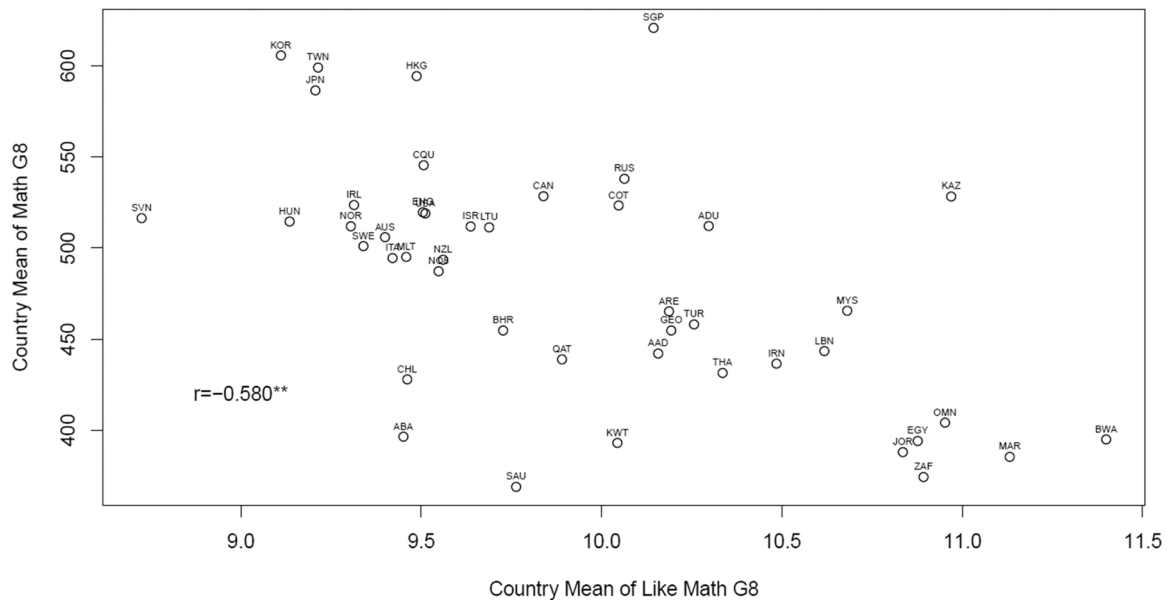


Fig. 2. Country Aggregated Correlation between Students Like Learning Mathematics and Mathematics Performance for Eighth Graders.

(the fit indices for metric model are very close to configural model, and the scalar model performs quite well), therefore, correlation analysis or regression analysis could be conducted using these scales.

3.2. Country level correlation analysis between students' interest and math and science achievement

In this section, the detailed description of the country level correlation results is presented for *Students Like Learning Mathematics (Fourth Graders)* and then results for the other seven scales are discussed together.

The results for *Students Like Learning Mathematics (Fourth Graders)* show that the student level correlations are positive for all countries, suggesting that there is a positive association between *Students Like Learning Mathematics* and mathematics performance at individual level for fourth graders. However, the country aggregated correlation shows a

negative association ($r_{\text{cnt level}} = -0.621, p < .01$). As shown in Fig. 1, East Asian education systems including Singapore, HongKong, Korea, Taiwan, and Japan all appear at the upper left corner, indicating that the fourth graders in these countries tend to like mathematics less despite their high performance. Since also some researchers found that students in Asian countries have higher anxiety because of high stake tests (Amorin & Lam, 2013) and in general lower self-confidence and self-esteem (Ho, E., 2007), researchers assumed that these correlations could be attributed to the East Asian countries participating in TIMSS. Consequently, we also calculated the correlations for non-East Asian countries only. After excluding the East Asian countries from the figure, a negative country level correlation is still obvious ($r_{\text{cnt level}} = -0.538, p < .01$), suggesting that this educational paradox is a global phenomenon and cannot be attributed to cultural differences between East Asian and non-East Asian countries.

The same procedure was applied to the other seven scales and similar

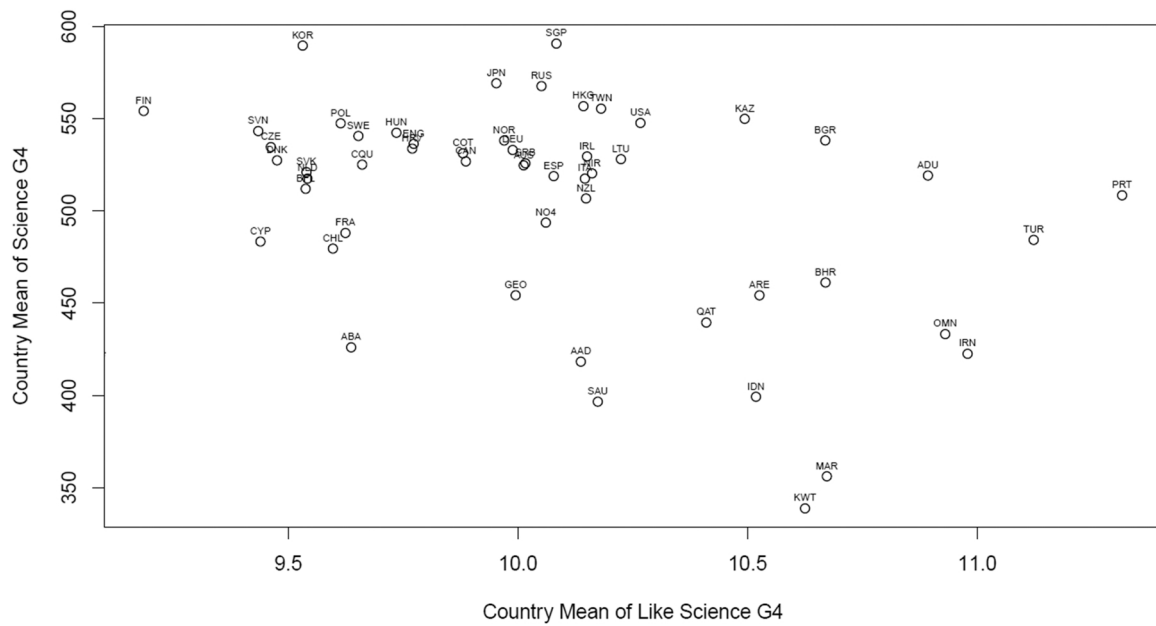


Fig. 3. Country Aggregated Correlation between Students Like Learning Science and Science Performance for Fourth Graders.

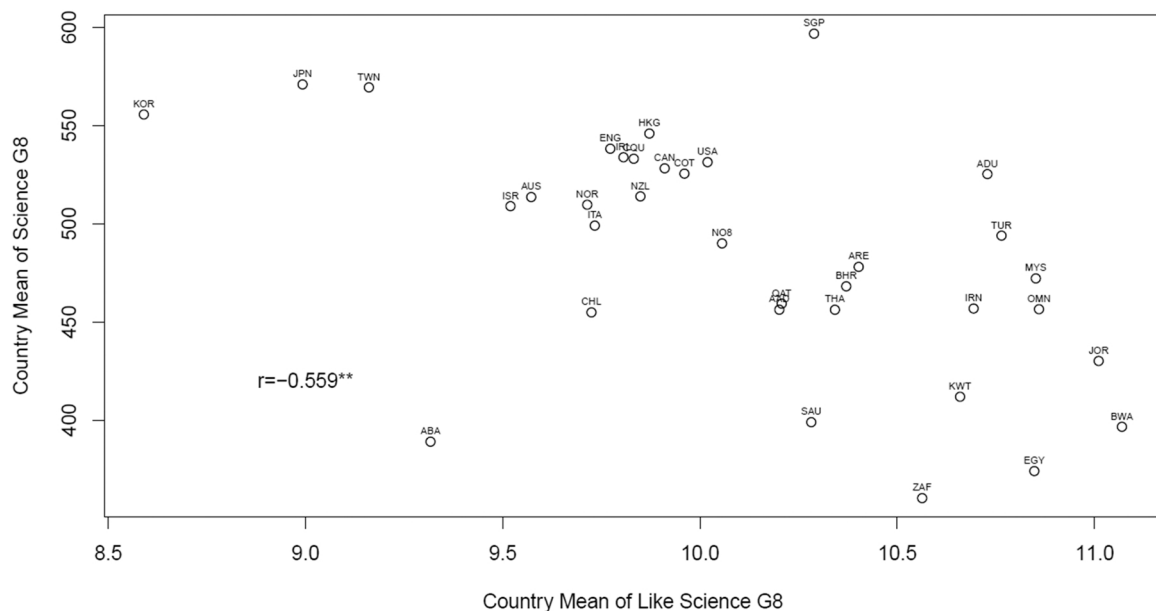


Fig. 4. Country Aggregated Correlation between Students Like Learning Science and Science Performance for Eighth Graders.

results were found. The student level correlation coefficients between the seven scales and the achievement are positive for almost all countries,¹ suggesting that there is a positive association between students' self-concept and interest in mathematics/science and mathematics/science performance at individual level for both fourth and eighth graders. However, the country aggregated correlations are all negative. As shown from Fig. 2 to Fig. 8, East Asian education systems including Singapore, HongKong, Korea, Taiwan, and Japan all appear at the upper left corner, indicating that the students in these countries tend to have low self-

concept and academic interest despite their high performance.

3.3. Class level correlation analysis between non-cognitive scales and math and science achievement

3.3.1. Students like learning mathematics for fourth graders (class level)

At individual level, there is a positive association between fourth graders' *Students Like Learning Mathematics* and their mathematics performance for the majority of classes. With regard to the class aggregated correlation, five countries/benchmarking participants including Belgium (Flemish), New Zealand, Serbia, Slovak Republic and Argentina, Buenos Aires show statistically significant negative correlation, indicating that for these countries, fourth graders in classes with high mathematic performance tend to have relatively lower appreciation of learning mathematics. Fig. 9 presents the class aggregated

¹ The student level correlation coefficient between Students Like Science and fourth graders' mathematics achievement for Sweden is -0.02 ($p > .05$); The student level correlation coefficient between Students Confidence in Science and fourth graders' science achievement for Malaysia is -0.16 ($p < .05$).

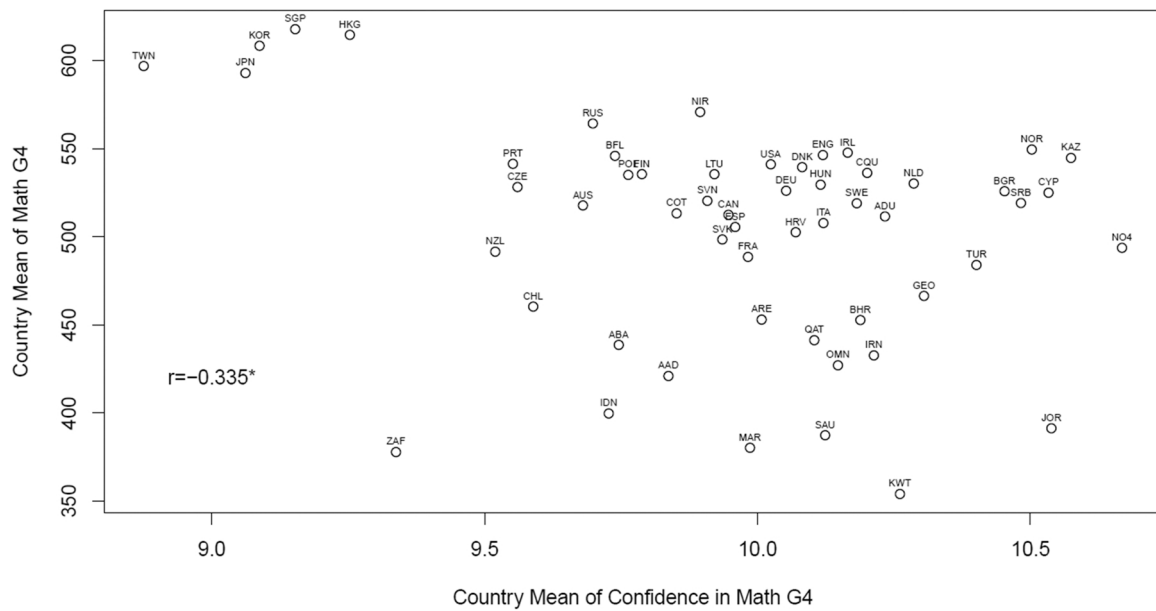


Fig. 5. Country Aggregated Correlation between Students Confidence in Mathematics and Mathematics Performance for Fourth Graders.

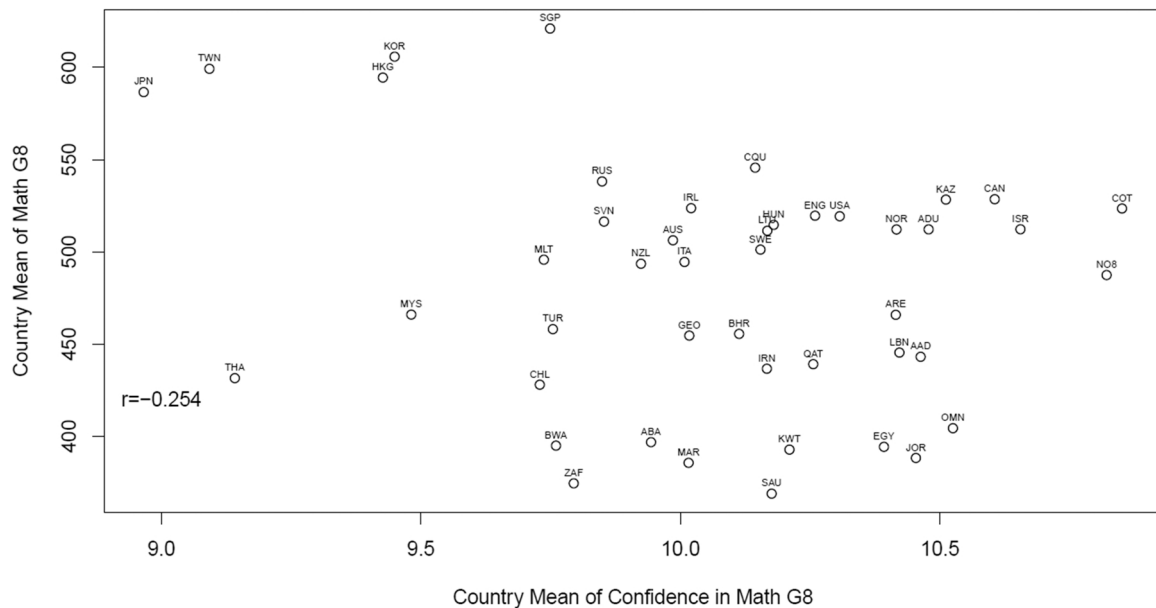


Fig. 6. Country Aggregated Correlation between Students Confidence in Mathematics and Mathematics Performance for Eighth Graders.

correlation between fourth graders' *Students Like Learning Mathematics* and their mathematics performance for the above-mentioned countries.

The same procedure was applied to other scales and similar results were found. At individual level, there is a positive association between the non-cognitive scales and the students' performance for the majority of classes. With regard to the class aggregated correlation, Botswana and South Africa show statistically significant negative correlation between eighth graders' *Students Like Learning Mathematics* and their mathematics performance; Serbia and Sweden show statistically significant negative correlation between fourth graders' *Students Like Learning Science* and their science performance, indicating that for Botswana and South Africa, eighth graders in classes with high mathematics performance tend to have relatively lower appreciation of learning mathematics; for Serbia and Sweden, fourth graders in classes with high science performance tend to have relatively lower appreciation of learning science. For self-concept scales, the class level aggregated

correlation are all positive for all countries. Thus, there seems no paradox between self-concept and achievement. Fig. 10 and Fig. 11 present the class aggregated correlation between the three scales and students' performance for the above-mentioned countries.

In summary, the educational paradox between non-cognitive scales (students' ASC and academic interest) and achievement exists at country level. At class level, there are more countries exhibiting such paradox for mathematics than science and for fourth graders than eighth graders. And no paradox for self-concept scales. However, it is noted that these conclusions are only based on descriptive rather than inferential statistics.

4. Discussion

The paradoxical relationship between students' non-cognitive scales (e.g., students' ASC and interest discussed in this paper) and their

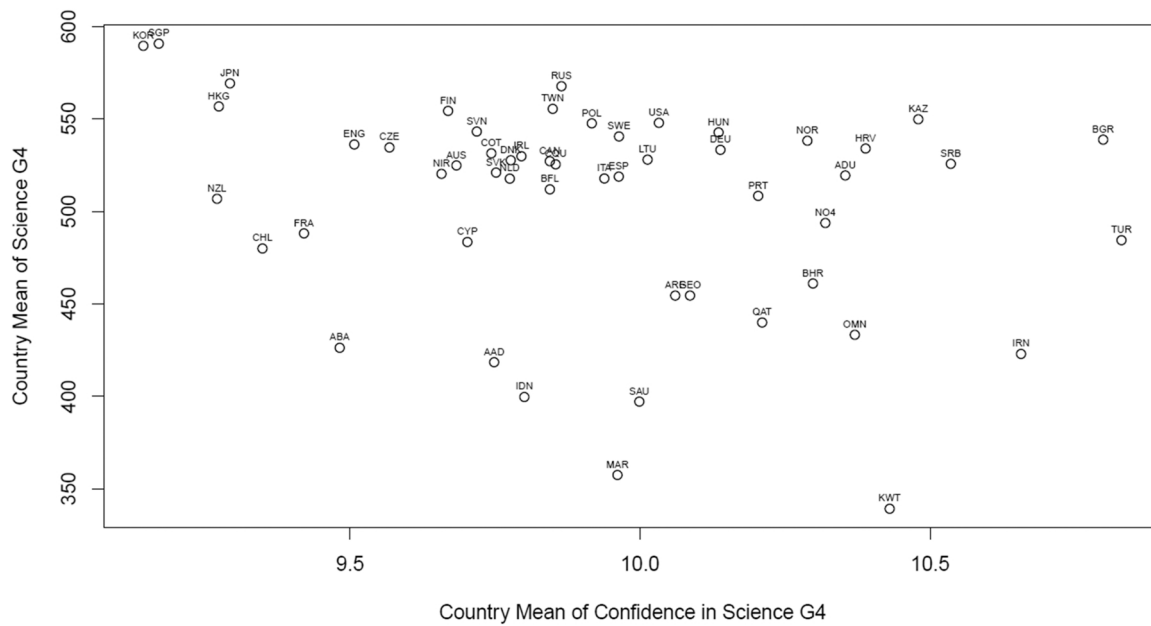


Fig. 7. Country Aggregated Correlation between Students Confidence in Science and Science Performance for Fourth Graders.

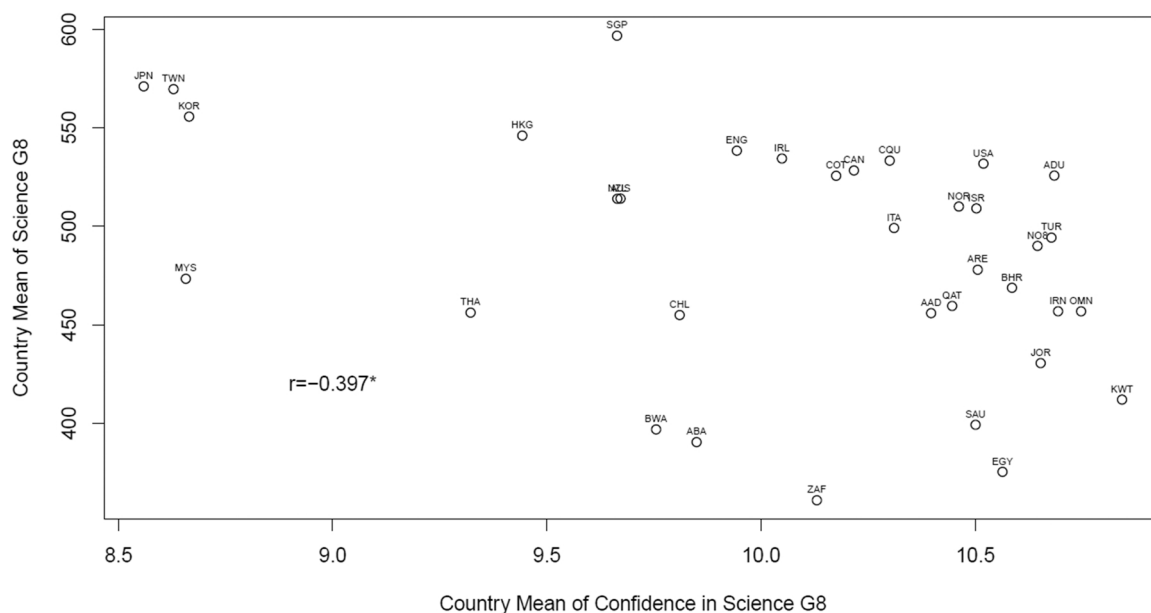


Fig. 8. Country Aggregated Correlation between Students Confidence in Science and Science Performance for Eighth Graders.

academic performance has been the interest of educators, researchers, and policy makers for a long period of time. People are trying to figure out to what extent such paradox exists and what are the reasons behind such paradox. Despite the accumulating literature on this topic, more studies are still needed to either complement what is missing in the literature or cross-validate what has been studied. This study aims to serve the first purpose. As mentioned in the introduction, few studies have ever checked the measurement invariance of the non-cognitive scales before analyzing their association with students' academic performance, which likely jeopardizes the validity of the statement that such paradox exists at the first place if the non-cognitive scales are not comparable among different groups. Our study fills this gap by checking two important non-cognitive scales: students' confidence and interest in the subject matter. Our results suggest that all the non-cognitive scales are measurement invariant across educational systems. However, it is

noted that the self-concept scales in this study includes more items into the scales compared with previous studies and some residual correlations were added to improve the performance of the self-concept scale before proceeding with measurement invariance analysis. It is suggested that researchers should agree on what items should be included to measure students' ASC. So far, the ASC scale in the field of paradox studies is not measured consistently.

The findings related to students' interest is more promising since this scale performs relatively better. Based on correlational analysis, some potential research questions emerge such as whether the paradox is more obvious for younger students than older students and for mathematics than science. In addition, based on what was found in our study, further research is needed to explore country specific explanations for the paradox where it exists.

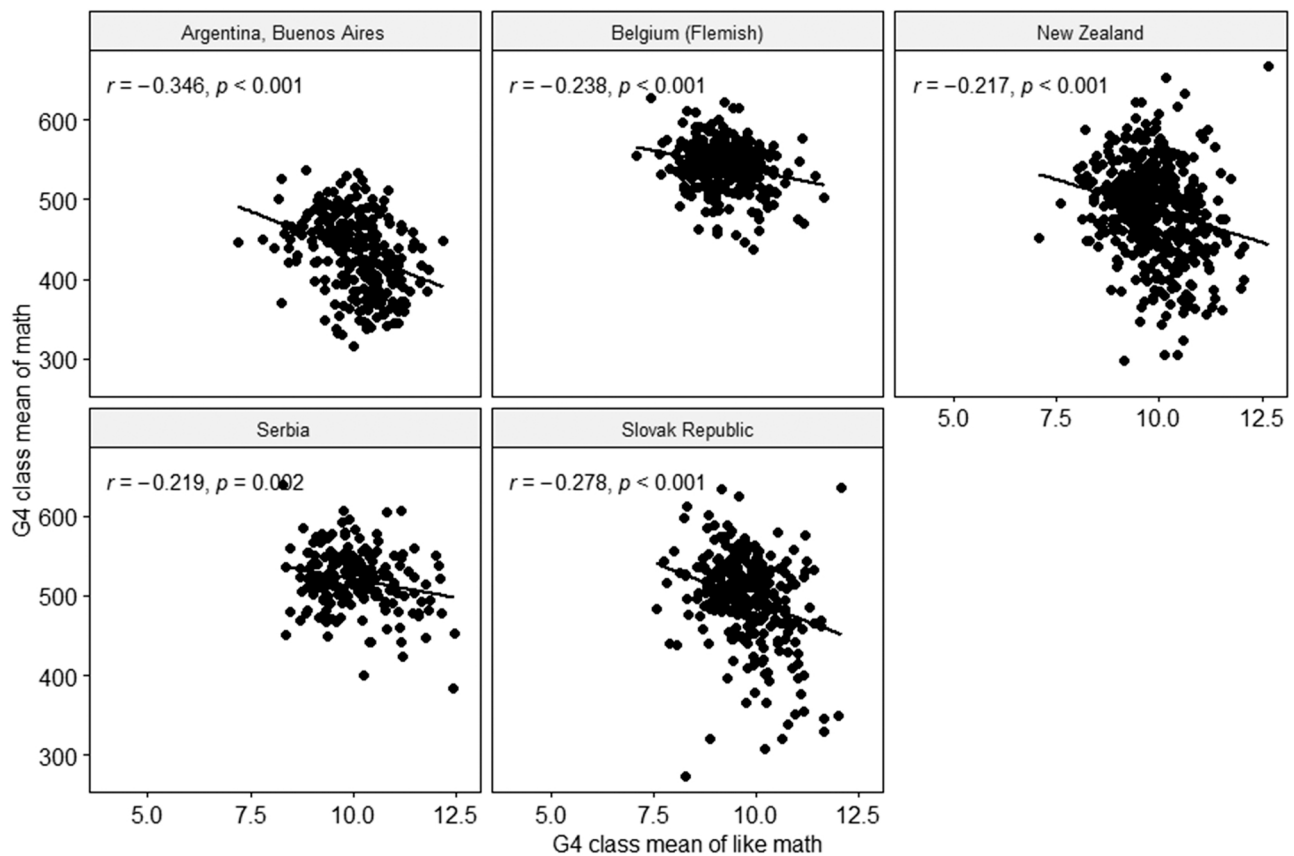


Fig. 9. Class Level Correlation between Students Like Learning Mathematics and Mathematics Performance for Fourth Graders.

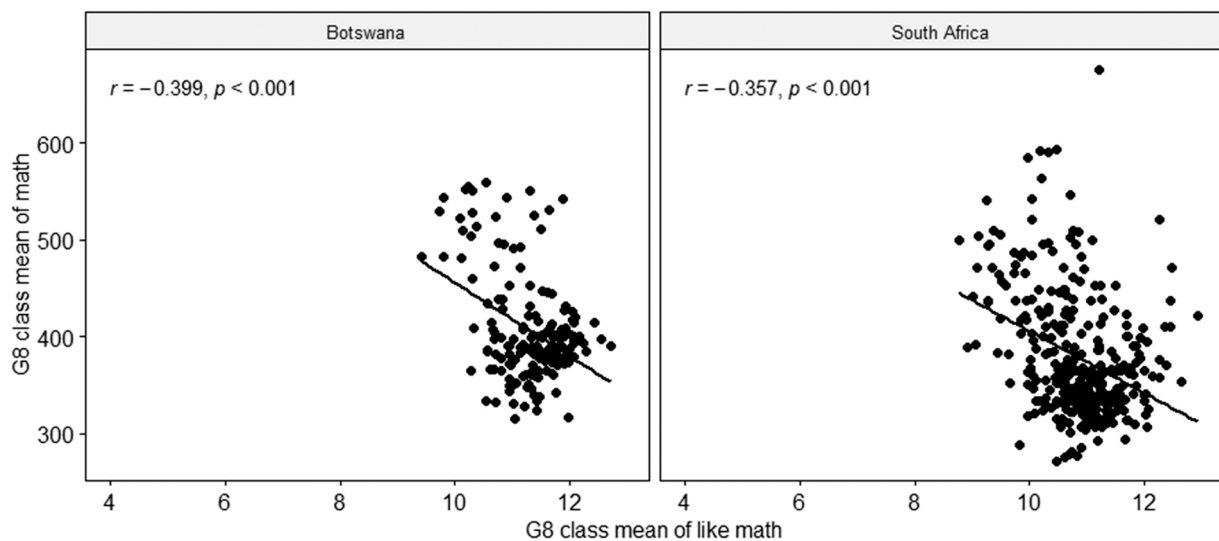


Fig. 10. Class Level Correlation between Students Like Learning Mathematics and Mathematics Performance for Eighth Graders.

5. Limitation and future studies

As mentioned before, except for the measurement invariance test of the two important non-cognitive variables using most recent TIMSS data, the remaining part of the study is correlational analysis, which limits our capacity to make any statistical conclusions. It is mostly due to the fact that the variation across different levels (individual level, class level, and country level) was not partialled out and analysed separately; MLM should be adopted to distinguish within variation and between

variation and will thus produce more accurate results.

Since we found the evidence that there is paradoxical relationship between students' non-cognitive scales and their mathematics/science achievement, it would be very interesting to investigate the differences in mathematics and science curricula. Will a curriculum analysis reveal that high achieving countries have demanding curricula, which negatively affect students' enjoyment in learning and their self-concept? Is teaching in higher achieving countries more strict or are there more drills included in teaching that create a positive impact on students'

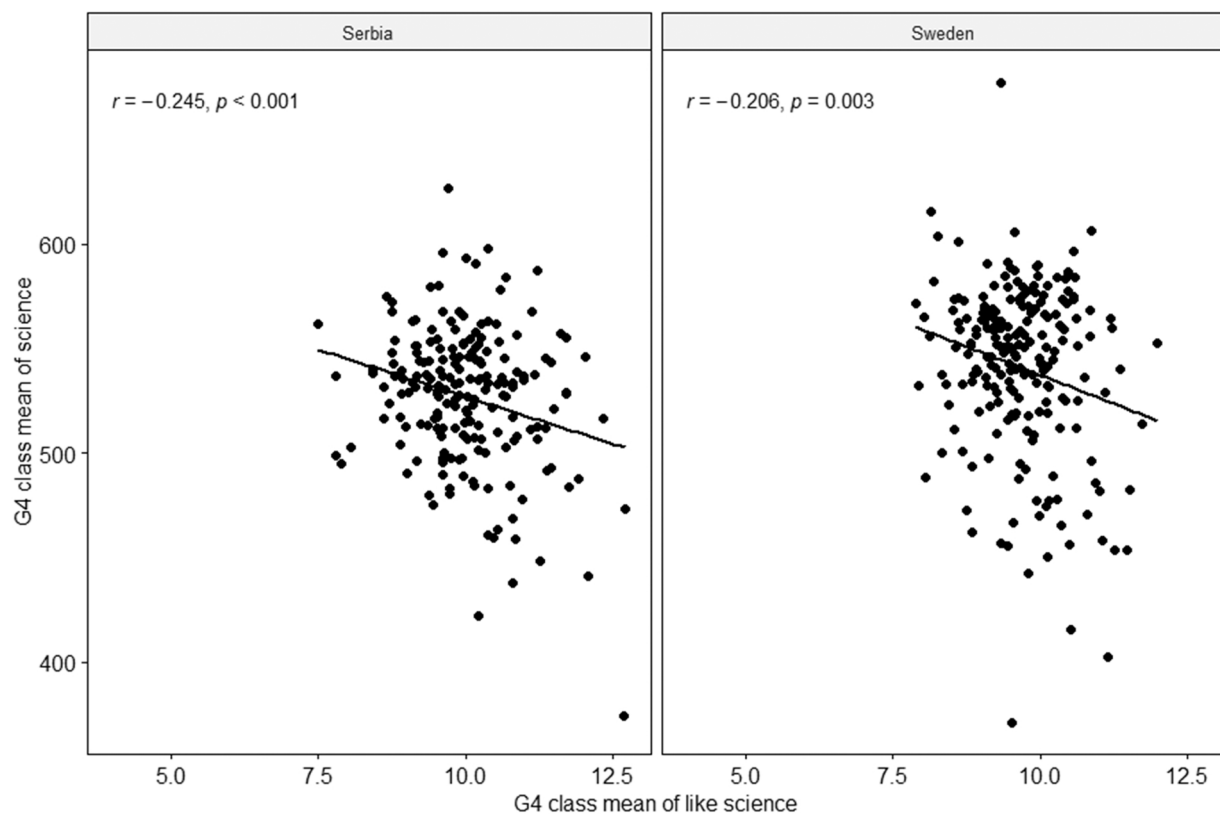


Fig. 11. Class Level Correlation between Students Like Learning Science and Science Performance for Fourth Graders.

achievement and a negative impact on enjoyment in learning and their self-concept? Further studies in this area are important to find optimal teaching and learning policies that lead to high student achievement and enjoyment of the subject at the same time.

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